



SAMUEL RAMOS VARELA – Machine Learning Engineer

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Summary – Mid-level Machine Learning Engineer with 3 years of experience in multimodal LLMs across pre/post training, optimization and evaluation. Currently focused on compression and deployment performance across CPU, GPU and TPU in HPC environments.

Personally fascinated by music production, graphic design, 3D point-cloud graphics and the beautiful unpredictability of fractals and chaos theory.

PROFESSIONAL EXPERIENCE

Machine Learning Engineer – MULTIVERSE COMPUTING

Madrid, Spain | December 2025 - Present

Compression of YOLO26 by 40-60%.

- Owned profiling, compression, healing and evaluation of YOLO26 at a 40-60% compression rate.
- Structured pruning, SVD and quantum methods. Healing with COCO and client use-case specific datasets.
- Performance and speed evaluation on CPU, edge and HPC. Inference optimization with TensorRT and ONNX.
- 40-60% compression. 2x throughput/FPS, 0.5x memory overhead with a minimal 2% impact on accuracy.

Compression of MOLMO2 by 65%.

- Owned the compression, healing and evaluation of Molmo2, a VLM for video understanding.
- Successfully pivoted from Molmo-7B to Molmo2-7B, released mid-project.
- Healing and evaluation with multimodal multitask benchmarks. Distributed workloads in HPC (MN5) and Skypilot.
- 65% compression. 3x token throughput, 3x latency speedup. Virtually no accuracy drop.

Company-wide model cost evaluation pipeline.

- Owned automation for comparing client model costs against CompactifAI API models.
- Real-world scenario benchmarking of throughput, concurrency, TTFT, etc. with llmperf and vllm.
- Estimated savings of ~240h and ~10.000€ per compression project.

Cross-functional Leadership and Ownership.

- Mentored junior ML Engineers on delivery-oriented experimental workflows and Git best practices.
- Acted as Project Owner and Technical Lead across several simultaneous projects.
- Owned client relationships with Fortune 500 strategic partners spanning IT, Energy, Finance and Defense.

Machine Learning Engineer – ASTOUND

Madrid, Spain | November 2024 - December 2025

ASSIST: Modular multimodal agentic pipeline.

- Applied ML leadership in a strategic partnership with “Thyssen” and “Colecciones Reales” museums.
- Modeled museum-like experience through a real-time multimodal agentic system: LLMs, VLMs, RAG, KG, ASR.
- LangGraph asynchronous architecture connected with FastAPI and vLLM, deployed in Azure.
- The software was sold for +70,000€ to a Spanish player in the museum audioguide space.

RAG and Knowledge Graph ingestion pipeline.

- Developed a reusable pipeline to ingest multimodal museum corpuses into graph (Neo4j) and vector DBs (qdrant).
- Automated relationship extraction and synthetic QA dataset generation.
- Fine tuning of the RAG VLM with LoRA and KD. LLM-as-a-judge evaluation.
- Curators reported saving ~12 hours of workload per new artwork and ~80 hours per thematic itinerary design.

User attention modeling pipeline.

- VLM layer for modeling short-term attention, LLM layer for aggregating long-term attention.
- RAG-inspired layer for zero-shot artwork detection (YOLOE) and identification (hybrid RAG with CLIP + LightGlue).
- Dynamic recommendation of artworks based on the user's attention.

Machine Learning Engineer – ITG TECHNOLOGY CENTER

Coruña, Spain | February 2023 - November 2023

NLP contextualisation toolkit.

- Engineered components for SER, intent detection, entity linking and semantic enrichment.
- Fine-tuned early Transformer and embedding models: BERT, RoBERTa, Word2Vec.

Computer Vision QC in manufacturing lines.

- Quality Control automation in packaging lines for Leyma. Construction of a dataset of anomalies and faults.
- Real time (30fps) edge inference on a Jetson with a finetuned YOLOv8.
- 16x more fault detections than the previous QC pipeline with minimal human monitoring.

LiDAR object detection.

- Detection of obstacles (bridge, dam, weir, etc) in rivers with LiDAR 3D scans and GIS, satellite and DEM data.
- Developed ensemble model with CNNs and classical methods for detection and characterization.
- River characterization efforts cut down to 10% of its usual workload by automating many field measurements.

RELATED PUBLICATIONS

Context or Retrieval? Evaluating RAG Methods for Art and Museum QA Systems - IWSDS 2025 - [ACL](#)

SER using multimodal LLMs and quality-controlled TTS-based data augmentation for Iberian languages - Computer Speech & Language Journal, July 2026 - [ScienceDirect](#)

ASSIST: A Multi-Agentic Framework for HCI in Cultural Heritage Settings - SIMBIG 2025 - [Springer](#)

ORCHESTRA: AI-Driven Microservices Architecture to Create Personalized Experiences - IWSDS 2026 - [ACL](#)

PATENTS

ArtMIRA: Art Museum Intelligent Recognition Assistant - *Software Intellectual Property Registration (No. M-002295/2026)* – Intellectual Property Registry of Madrid. Registered Feb 2026.

Multimodal-AIC: Module for Automatic Analysis of Attention, Intentionality and Contextualization (AIC) using Multimodal Information - *Software Intellectual Property Registration (Ref: 09/410084.9/26)* – Intellectual Property Registry of Madrid. Registered Feb 2026.

TECHNICAL SKILLS

- **Core ML:** Python, PyTorch, TensorFlow, Hugging Face, CUDA, OpenCV, FastAPI, docker swarm, JAX, Transformers
- **LLMs & Agentic AI:** vLLM, LangChain, LlamaIndex, LangGraph, CrewAI, n8n, MCP, SQL, RAG, qdrant, ChromaDB, Knowledge Graphs, Neo4j, SFT, DPO, Early Fusion, MoE (Mixture of Experts), LLM-as-a-Judge, Knowledge Distillation, Evaluation (RAGAS, Deepeval), Latency/Throughput Benchmarking
- **Cloud & HPC:** AWS, Azure, GCP, MN5 Supercomputer, TensorRT, ONNX, Docker, MLflow, Coder, SkyPilot
- **Project Management:** Agile (Scrum, Kanban), CI/CD, Collaborative Code Management (Git), Project Ownership, Technical Leadership, Mentorship
- **Languages:** English (Native), Spanish (Native), Galician (Native), German (Fluent)

EDUCATION

(MSc) Master of Science in Signal Processing and Machine Learning for Big Data - 7.5/10

UPM – Universidad Politécnica de Madrid | Feb 2025 - Feb 2027

(BSc) Bachelor of Science in Computer Science – Artificial Intelligence Specialization - 8/10

UDC – University of A Coruña | Sep 2019 - Jun 2024

Exchange Program in Computer Science - 9.5/10

KU – Korea University – KU | Aug 2021 - Feb 2022